Post-Training Qwen3-1.7B for Verifiable Reasoning and Alignment

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Abstract

Large language models remain fragile under verifiable benchmark evaluation, where malformed, verbose, or unsafe outputs are automatically penalised. We post-train Qwen3-1.7B on four benchmarks: mathematical reasoning, safety, general knowledge, and multilinguality using benchmark-specific LoRA adapters and evaluating SFT, DPO, GRPO, chain-of-thought distillation, and model merging for the final group model. The central design principle is format-aligned learning: all models produce a final answer inside `\boxed{}`, matching the official evaluator. Our main finding is that clean SFT with curated, format-aligned data consistently outperformed aggressive refinements: DPO, GRPO, and thinking-mode training inflated generation length and degraded boxed-answer compliance. Overall, data quality, task alignment, and strict output formatting mattered more than dataset size or adding longer reasoning.

1 Introduction

Large language models can produce fluent answers while failing strict benchmark evaluation due to malformed outputs, over-explanation, or unsafe responses; a gap that general fine-tuning worsens rather than resolves. This is especially critical in verifiable multiple-choice settings, where a missing or malformed final answer is automatically penalised. We study this across four capabilities. For mathematical reasoning, MATH [1] motivates our math adapter because it evaluates challenging competition-level problem solving where precise final answers are essential. For general knowledge, MMLU [2] motivates our knowledge adapter because it measures broad academic and professional understanding across many subjects. For multilinguality, Global-MMLU [3] motivates our multilingual adapter because it exposes how English-centric pretraining can degrade reasoning across languages and cultures. For safety, SafetyBench motivates the adapter as it evaluates whether models can select safe responses across diverse risk categories [4]. Existing post-training methods can improve task-specific performance, but they also introduce important limitations. SFT [5] may overfit to dataset style, DPO [6]/GRPO [7] can destabilize short-answer formats, CoT training increases verbosity, and multi-task training causes cross-domain interference. To address these issues, we post-train Qwen3-1.7B using four benchmark-specific LoRA adapters [8], and build a group model by merging them as an alternative to direct multi-task training, inspired by TIES-Merging [9]. Our experiments show that SFT with curated, format-aligned data is the most reliable strategy, while DPO and GRPO refinements were less stable; enabling thinking often increased generation length and reduced boxed-answer compliance.

2 Method

We post-train Qwen3-1.7B into four benchmark-specific models and one group model. All benchmark-specific models use LoRA for efficient adaptation with a limited number of trainable parameters. All models were trained and evaluated with prompt templates enforcing a final answer inside `\boxed`. For Safety, General Knowledge, and Multilinguality, the task is formulated as multiple-choice question answering: given a question and 2–20 candidate options, the model may reason before answering but must finish with exactly one option letter inside the box. In contrast, Math requires a numerical or short symbolic final answer.

Mathematical Reasoning We trained LoRA SFT models for exact-answer generation, scaling from small curated reasoning mixtures to larger long-context math mixtures. We also tested worked-solution supervision and GRPO refinement from SFT. The final strategy used format-aligned SFT, because failures involved both incorrect reasoning and malformed boxed answers. We selected checkpoints with pass@8, which rewards correct and correctly formatted answers across multiple sampled attempts.

General Knowledge We aimed to improve academic multiple-choice accuracy, using MMLU auxiliary train as the main starting point and adding harder sources for wider coverage. Motivated by the base model's malformed and overly verbose outputs, we trained no-thinking SFT checkpoints with an MCQ prompt to enforce option selection and reliable formatting; the General Knowledge prompt variants are reported in Appendix B.2. Inspired by teacher-student distillation, we also tested Qwen3-8B CoT traces filtered for correctness. Finally, we applied GRPO on hard examples with rewards for correct boxed answers and penalties for missing boxes. The final model enables thinking at inference time on the best SFT checkpoint.

Multilinguality targets Italian, Spanish, Chinese, Hindi, and Russian MCQs. We use CoT distillation from Qwen3-

14B, the biggest teacher fitting inside a single A100 GPU and using the same tokenizer [10], retaining only correctly answered and formatted traces, then apply SFT on a disjoint training partition for format alignment. We experimented with GRPO to improve reasoning efficiency. We selected the DISTIL-SFT-GRPO pipeline, which has the best accuracy on our validation set.

Safety is formulated as safest-option selection: given multiple candidate responses , the model must choose the least harmful or most responsible option. We use SFT on curated safety, ethics, bias, social-reasoning, and toxicity mixtures, then compare it with DPO and thinking/CoT variants. Since safety evaluation penalizes both unsafe choices and malformed answers, the final strategy favors direct format-aligned SFT, which proved more stable than preference optimization or reasoning-heavy variants.

Group Model combines the four benchmark-specific models. We compared two strategies: multi-task fine-tuning on the combined data and post-hoc merging of independently trained LoRA adapters. Multi-task distillation from Qwen3-14B then SFT-GRPO disjoint data from mixed subsets of all domains did not improve over the thinking baseline, suggesting cross-domain interference. We also attempted TIES-merging which was better but differences in adapter ranks and training scales caused trimming. We therefore adopted a direct weighted linear merge of the four LoRA adapters. The merged model uses thinking mode on a unified prompt template E.1 for both MCQ tasks and open-ended Math while enforcing the same final `\boxed` format across benchmarks.

3 Experimental Setup

3.1 Datasets

For all domains, we used training splits for fine-tuning and kept evaluation splits separate to reduce contamination.

For Math, we used OpenMathReasoning [11], Numina-Math CoT [12], competition-style problem collections, and the math subset of AceReason-1.1-SFT [13, 14]. These sources provide worked solutions and final answers across algebra, number theory, combinatorics, geometry, probability, and other competition domains. We built smaller curated mixtures for format alignment and larger mixtures of up to 100k samples for long-context training, with more medium/long-form reasoning traces.

For General Knowledge, we first used standard academic and science MCQ datasets, including MMLU auxiliary [2], SciQ [15], OpenBookQA [16], and ARC-Challenge [17], to establish the training pipeline. To meet hidden benchmark difficulty, we shifted to harder and broader mixtures using GPQA [18], SuperGPQA [19], larger MMLU auxiliary subsets [2], Quartz [20], and Nemotron-RL Knowledge MCQA [21]. For reasoning experiments, we also used Llama-Nemotron science examples with reasoning-style rationales [22]. Because these datasets had heterogeneous question, option, answer, and rationale formats, each required a dedicated converter.

For Multilinguality, we train on Italian, Spanish, Chinese, Hindi, and Russian MCQs spanning national exams, professional licensing, culture, and academic questions. Cross-lingual coverage comes from Global-MMLU [3], Kaleidoscope [23], and M3Exam [24]. Language-specific data included ITALIC [25], EXAMS [26], AYA-ES, HEAD-QA [27], MedExpQA-ES [28], exams-bolivia [29], Peruvian-exams [30, 31, 32], CMMLU [33], C-Eval [34], CMB [35], MMLU-Hindi [35], Indic-MMLU [36], Hindi-MCQ [37], and ruMMLU-Pro [38]. Per-source caps were applied to avoid dominance by any single language or dataset, targeting roughly 2.5k samples per language.

For Safety, we used heterogeneous safety and social-reasoning datasets converted into MCQ form. BeaverTails [39] provided harmful prompts with safe and unsafe responses; HH-RLHF [40] preference pairs were converted into two-option MCQs; SIQA [41], ETHICS [42], Civil Comments [43], BBQ [44], SALAD [45], and WildGuardMix [46] covered social reasoning, ethical acceptability, toxicity, bias, harmful prompt classification, refusal detection, and safe response selection. Most Safety data used direct boxed answers, except in later thinking experiments.

Across domains, we converted all datasets to a shared chat format with a final `\boxed{}` answer. For MCQ tasks, we preserved variable option counts, randomized options when applicable, and required only the option letter inside the box; for Math, we required a parsable numerical or symbolic boxed answer. We removed malformed, duplicate, and validation-overlapping samples. Ethically, we note that safety datasets may encode annotator bias, multilingual data overrepresents higher-resource languages, and web-sourced math data may contain benchmark-adjacent duplicates, motivating filtering and overlap checks.

3.2 Evaluation

Math is evaluated with pass@8, while all other tasks use pass@1 accuracy. We extract the final answer and compare it to the gold label: a letter for MCQs or a numerical expression for Math. Missing, malformed, or unboxed outputs are marked incorrect and counted as parse failures.

For Math, we used a fixed local validation set combining MATH-500 [47], historical AIME problems [48, 49], recent AIME sets [50, 51, 52], and additional validation samples. Its full composition is reported in Appendix A.1. The base Qwen3-1.7B model achieved 24.44% pass@8, serving as the main baseline. We also compare against intermediate checkpoints corresponding to curated SFT, GRPO refinement, all-solution supervision, and long-context SFT to isolate the effect of each design choice.

For General Knowledge evaluation, we used a fixed 2,010-example local validation set drawn from MMLU [2] (455), MMLU-Pro [53] (845), SuperGPQA [19] (400), Nemotron-RL Knowledge MCQA [21] (300), and the provided course validation samples (10) (see Figure 2 in Appendix B.1). This fixed mixture covers standard academic and harder reasoning-intensive MCQs across science, law, medicine, humanities, and social sciences, enabling fair local comparisons across experiments. We also used the CI validation score as an external signal, selecting models based on both local and CI performance to reduce

overfitting to our local set. As baselines, we evaluated Qwen3-1.7B with the non-thinking and thinking prompts reported in Appendix B.2. The baselines motivated our final strategy: the non-thinking model had poor boxed-answer compliance and only ∼8% local accuracy, while thinking improved accuracy to 48.4% but remained imperfectly formatted. We therefore used SFT for reliable formatting and enabled thinking at inference time.

For multilinguality, evaluation is conducted on 1,500 held-out examples (300 per language), sampled from test or validation splits, weighted by nativity and subject alignment. Sources split into two tiers. *unseen* contains INCLUDE (natively-authored regional/civic/cultural mcqs for all languages)[54] and MILU [55] (Indian exams, Hindi), whose benchmarks were never used during training. *seen* contains EXAMS, HEAD-QA, CMMLU, C-Eval, Indic-MMLU, and Global-MMLU [3], whose train splits were used for training. Figure 7 shows dataset composition for validation. Qwen3-1.7B with thinking enabled (53.9%) is the baseline (Appendix C.2), a strong bar given Qwen3's multilingual pretraining [56].

For Safety, we built a held-out evaluation set of 935 examples across eight task types , covering toxicity comparison [39], social norms [41], bias detection [44], ethics identification and judgement [42], SafetyBench MCQs [4], and official course samples; the full breakdown is shown in (Figure 9). We report local clean accuracy on this set and CI pass@1 when available. The non-thinking baseline Qwen3-1.7B achieved only 16.47% accuracy, with many responses partially correct under fallback extraction but non-compliant with the required format.

Group model evaluation uses a unified 2k benchmark samples, 500 from each validation set described above. We report both overall and per-domain accuracy to detect the effects caused by combining the specialized adapters. As a baseline, we evaluated the base Qwen3-1.7B model on the same set using a minimal boxed-answer prompt with thinking enabled; the prompt is reported in Appendix E.1. It achieved 51.95% overall accuracy.

3.3 Implementation Details

All models are fine-tuned from Qwen3-1.7B using LoRA on A100-40GB GPUs with AdamW and dropout 0.05 . All domains were trained for one epoch, except Math where we explored 0.5, 0.7, and 1 epoch variants. GRPO runs used batch 2 with 8 generations per question at temperatures 0.9-1.1, with top-p = 0.95 and top-k = 20. Table 1 summarises the per-domain hyperparameters.

Domain	Setup	r/α	LR	Seq.	Batch	Time
Safety	SFT (direct)	16/32	$1–2\times10^{-5}$	2048	16	∼2h
	SFT (CoT)	16/32	$1–2\times10^{-5}$	3072	16	∼4h
Gen. Knowl.	SFT (CoT)	16/32	$1–5\times10^{-5}$	1024	16	∼2h
	Thinking SFT	32/64	$1–5\times10^{-5}$	4096	16	∼4h
Multilinguality	Distillation SFT	64/64	5×10^{-5}	4096	16	∼10h
	GRPO	64/64	5×10^{-5}	512	16	∼2h
Math	SFT	16–96	–	2k–4k	16	∼2h
	long-ctx SFT	64/128	7.5×10^{-5}	16k	32	∼7h

Table 1: Hyperparameters per domain and setup.

4 Results

4.1 Main Results

Table 2 reports the main local evaluation results, with CI scores shown separately for external validation. All individual models substantially improve over their baselines, reaching 66.79% on Math, 53.23% on General Knowledge, 55.9% on Multilinguality, and 77.22% on Safety. The best group model reaches 57.9% average local accuracy, with 47.8%, 59.2%, 49.2%, and 75.2% on the four domains respectively. This indicates competitive merged performance, although some cross-domain interference remains compared with specialized models. Detailed per-domain tables are reported in the appendix for each benchmark.

Model	Math	Gen. Knowl.	Multilingual	Safety	Avg.
Individual baseline	24.44%	48.4%	53.9%	16.47%	
Best individual model	66.79%	53.23%	55.9%	77.22%	
Best individual model CI	54%	46%	60%	84%	
Group baseline	32.2%	53.2%	46.2%	76.2%	51.9%
Best group model (local)	**47.8%**	**59.2%**	**49.2%**	**75.2%**	**57.9%**
Best group model CI	**49%**	**47%**	**58%**	**85%**	**59.5%**

Table 2: Comparison of individual and group models across all four benchmarks.

4.2 Analysis

Mathematical reasoning. Ablations show a clear progression from format alignment to long-context reasoning. Detailed checkpoint results and the full trajectory are reported in the appendix (Table 4, Figure 1). The base model reached 24.44% pass@8, with many failures caused by missing or unparseable boxed answers. Early SFT enforced valid outputs to mitigate the issue, but performance plateaued with small curated mixtures. GRPO refinement and all-solution supervision did not improve further. GRPO likely suffered from a sparse reward signal under strict format evalutation, while all-solution training increased reasoning length without guaranteeing final answers. The largest gain came from AceReason-style mixtures and 16k-context training, which better matched the long multi-step structure of competition-style solutions. The best pushed checkpoint reached 66.79% pass@8 on the local validation set and 54% on CI. Table 5 shows that long-context training improved especially on MATH-500, historical AIME-style problems, and AIME 2025, while AIME validation and course samples remained difficult. Overall, Math performance depended more on data quality, long-context alignment, and strict final-answer formatting than on additional optimization stages or longer traces.

General Knowledge. The main challenge was balancing accuracy with boxed-answer extraction. The non-thinking base model reached only ∼8% local accuracy due to malformed outputs, while enabling thinking raised accuracy to 48.4%, revealing useful reasoning ability but weak format reliability. We therefore used non-thinking SFT for option selection and enabled thinking only at inference time. SFT improved local accuracy, but larger mixtures did not always transfer to CI. In Table 6, the best checkpoint was Exp. 1, trained on a ∼60k-example mixture of MMLU auxiliary,

Nemotron-RL Knowledge MCQA, GPQA, and SuperGPQA. It reached 47.62% local and 44% CI, whereas the larger 126k-example Exp. 3 reached 49.90% locally but only 37% on CI. Smaller and intermediate ablations are reported in Appendix B.4, Table 7. Reasoning-oriented refinements were less reliable. GRPO had negligible impact (Table 8), and Qwen3-8B CoT distillation reached 50.15% locally and 39% on CI, but longer DeepSeek-R1-style rationales reduced local accuracy by 1% due to verbosity and wrong formatting (Table 9). The strongest setting enabled thinking at inference time on the selected SFT checkpoint, reaching 53.23% local accuracy and 45% CI. Overall, SFT-learned format discipline plus inference-time reasoning outperformed more data, GRPO, and longer CoT traces. The final model performs best on Nemotron and MMLU, but remains weaker on harder sources such as MMLU-Pro and SuperGPQA (Appendix B.7).

Multilinguality The baseline achieved 53.9% accuracy from prompting alone (Table 10). Per-language analysis (Table 12) shows Chinese consistently outperforms other languages, while Hindi remains the weakest despite balanced training data, suggesting inherent base-model variation. Distillation did not improve over SFT, as the teacher's CoT traces added no further knowledge. GRPO yielded no accuracy gain, but Figure 8 shows reward shaping shortens generations without much reward loss at the end of training, reducing inference cost. Forcing the model to reason in the question's language proved counter-productive: despite Qwen3's multilingual pretraining [56], English consistently outperformed native-language reasoning even on culturally specific questions (Table 13) , due to the English dominance in LLM pretraining data [57]. Table 11 shows unseen accuracy exceeds seen by 3-4% across the top models, confirming that fine-tuning generalizes beyond exact training subjects. Overall, Distil-SFT-GRPO achieved the best balance of accuracy and generation efficiency.

Safety. Scaling dataset diversity mattered more than size (Table 14): moving from BeaverTails-only SFT to a broad mix of social, ethical, bias, and toxicity data drove the largest gain (67.7% $\rightarrow$ 77.22%), while scaling to 58.8k hurt performance as the BBQ-heavy mixture diluted the safety-reasoning signal. The per-category breakdown (Figure 10) reveals a clear asymmetry: toxicity comparison (92.0%) and bias detection (90.0%) are strongest, while ethics acceptability (37.5%) and ethics identification (63.0%) remain hardest. DPO (Appendix D.5) marginally improved clean accuracy but hurt CI (84% $\rightarrow$ 83%), and iterative DPO caused severe format collapse, confirming RL refinement's fragility on near-deterministic single-letter outputs. Thinking-mode and CoT variants (Appendix D.6) all underperformed direct SFT: even carefully distilled teacher traces (Claude Haiku 4.5, $\sim$85% keep rate, 18k examples) reached only 72.51% clean, as the longer reasoning format made optimization harder. Clean, diverse, format-aligned SFT remained the most robust strategy.

Group Model. The base Qwen3-1.7B achieved 51.9% on the group benchmark, proving strong out-of-the-box cross-domain capabilities. We first explored multi-task fine-tuning, which failed to match individual models performance due to cross-domain interference: direct multi-task SFT reached only 37.8% locally, while a Distillation-SFT-GRPO variant improved to 48.15% (Table 17). We also attempted adapter merging: TIES-merging reached 54.6% locally, below the linear merge, as rank and scale differences caused trimming to remove domain-critical parameters (Appendix E.3, Table 18). Direct linear merging preserved all updates and improved to 57.9% locally and 59.5% CI (Figure 11, Appendix E.4). Gains were uneven: Safety benefited most (76.0% local, 85.0% CI), its adapter producing the clearest separable delta; Math (47.8%, 49.0%), Multilinguality (49.2%, 57.0%), and General Knowledge (59.2%, 47.0%) showed more interference. Sub-source analysis (Figure 12) confirms that standard benchmarks were preserved while harder subsets (AIME, SuperGPQA, ethics acceptability) degraded, highlighting uneven sub-domain interference as the main limitation.

5 Ethical Considerations

Our work benefits researchers seeking reliable, evaluator-compatible small models. However, benchmark-oriented training has risks: safety datasets may reflect annotator bias, and optimizing for boxed answers may improve scores without ensuring open-ended robustness. These risks disproportionately affect marginalised groups, who are likely to be misrepresented in safety and multilingual training data. We mitigate them by treating scores as proxies, using held-out evaluation data, tracking malformed outputs, and reporting limitations. Since our multilingual experiments cover only five mid-to-high-resource languages, the conclusions may not transfer to low-resource settings. Still, the work remains useful because it shows how careful data curation and format alignment can improve reliability without relying only on larger models.

6 Conclusion

We post-trained Qwen3-1.7B across four benchmarks using LoRA-based SFT, distillation, and model merging. The central finding is that output-format compliance drives benchmark performance: models frequently failed not from lack of knowledge but from malformed or unboxed responses. Clean SFT with curated, format-aligned data consistently outperformed DPO, GRPO, and thinking-mode refinements, which in some cases inflated generation length and degraded boxed-answer compliance. The merged group model improved over the prompted baseline by 6 points overall (51.95% $\rightarrow$ 57.9%), while individual models reached 77.22% on Safety, 53.23% on General Knowledge, 66.79% on Math, and 55.9% on Multilinguality. Limitations include reliance on a single model family (Qwen3), coverage of only five mid-to-high resource languages, and evaluation restricted to multiple-choice formats that reward confident answers over calibrated reasoning. Future work should explore cross-family adapter merging, low-resource language transfer, and evaluation paradigms that capture open-ended reasoning robustness.

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A Math Appendix

This appendix reports the detailed Mathematical Reasoning experiments supporting the analysis subsection. Local scores are computed on the fixed 536-example validation set, while CI scores are reported when available. Local evaluation was used for controlled checkpoint comparisons, and CI was used as an additional signal for benchmark generalization.

A.1 Evaluation Set Composition

The fixed Math validation set contains 536 examples combining standard mathematical reasoning and competition-style problems. We kept this set unchanged across all Math experiments so that local pass@8 scores were directly comparable.

Source	Number of examples
MATH-500	200
Historical AIME-style problems	194
AIME validation	90
AIME 2025	29
AIME 2024	13
Course-provided validation samples	10
Total	536

Table 3: Composition of the fixed Math validation set used for local pass@8 evaluation.

A.2 Prompt Format

Math prompt format

Solve the following mathematical problem carefully. Use concise reasoning and finish with exactly one final answer inside `\boxed{}`. Do not put explanations or extra text inside the box.
Question: {problem_text}
Expected assistant output:

```
Reasoning steps ...
\boxed{final_answer}
```

A.3 Main SFT and Checkpoint Experiments

Table 4 summarizes the main Math ablations. Early checkpoints used a 2k generation budget and mainly tested format-aligned SFT, GRPO, and all-solution supervision. Long-context checkpoints used a 12k generation budget and evaluated the effect of larger AceReason-style reasoning mixtures.

Model	Main idea	Local pass@8	Count	CI
Base Qwen3-1.7B	Prompt only	24.44%	131/536	–
Early SFT	Small LoRA SFT mixture	29.29%	157/536	25%
Revised SFT	Revised LoRA SFT mixture	25.56%	137/536	–
Curated SFT	Curated reasoning mixture	30.60%	164/536	29%
Strong curated SFT	Best early SFT checkpoint	37.31%	200/536	39%
GRPO refinement	Reward refinement from SFT	36.38%	195/536	–
All-solution SFT	Complete solutions, $r = 96$	36.19%	194/536	–
50k long-context SFT	AceReason-style 50k mixture	65.30%	350/536	48%
100k long-context SFT	100k mixture, ckpt1600	64.55%	346/536	52%
100k long-context SFT	**100k mixture, ckpt2400**	**66.79%**	**358/536**	**54%**

Table 4: Detailed Math checkpoint ablations. Local pass@8 is evaluated on the fixed 536-example validation set. Early checkpoints use a 2k generation budget, while long-context checkpoints use a 12k budget.

These ablations show two main trends. First, format-aligned SFT substantially improved over the base model, but gains saturated with smaller curated mixtures. Second, the largest improvement came from changing the data and context regime: the 50k and 100k long-context SFT models more than doubled the base pass@8. GRPO and all-solution supervision were negative or neutral ablations, indicating that additional reasoning length or reward optimization did not help unless the model remained concise and consistently ended with a valid boxed answer.

A.4 Source-Level Breakdown

Table 5 compares the strongest long-context models by source. MATH-500 is the easiest subset for both models, while AIME validation and course-provided samples remain harder. The 100k long-context checkpoint improves especially on MATH-500, historical AIME-style problems, and AIME 2025.

Source	50k long-context SFT	100k long-context SFT
MATH-500	90.50%	91.50%
AIME 1983–2024	63.40%	64.43%
AIME validation	30.00%	30.00%
AIME 2025	31.03%	44.83%
AIME 2024	53.85%	53.85%
Course math	30.00%	30.00%
Overall	65.30%	66.79%

Table 5: Source-level Math validation breakdown for the strongest long-context models.

A.5 Ablation Trajectory

Figure 1 summarizes the full Math ablation trajectory. The first phase shows that early format-aligned SFT improves over the base model but remains limited with small curated mixtures. The second phase shows a plateau: GRPO refinement and all-solution supervision do not improve over the strongest curated SFT checkpoint, indicating that reward optimization or longer traces alone did not solve the extraction-based evaluation bottleneck. The third phase shows the main gain: long-context SFT on

AceReason-style mixtures raises local pass@8 from the mid-30% range to above 65%, with the best pushed checkpoint reaching 66.79%. This supports the main conclusion that Math performance was driven primarily by data quality, long-context alignment, and reliable final-answer formatting.

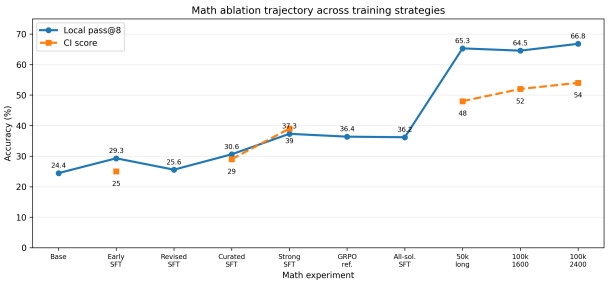


Figure 1: Math ablation trajectory across training strategies. Local pass@8 is measured on the fixed 536-example validation set; CI scores are shown when available.

B General Knowledge Appendix

This appendix reports the detailed General Knowledge experiments supporting the analysis subsection. All local scores are computed on the fixed 2,010-example validation set, while CI scores are reported on the course validation benchmark when available. Local evaluation was used for controlled comparisons across runs, and CI was used as an additional signal for hidden-benchmark generalization.

B.1 Evaluation set composition

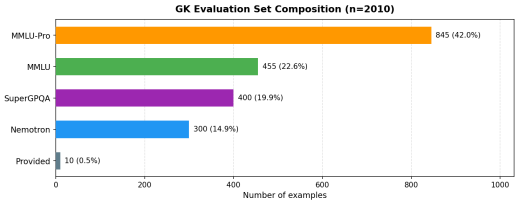


Figure 2: Composition of the 2010-example General Knowledge evaluation set

B.2 Prompt formats for thinking and no-thinking experiments

To compare direct-answer fine-tuning with reasoning-based fine-tuning, we used two different target formats. In the no-thinking setting, the model was instructed to directly output the boxed answer. In the thinking setting, the model was allowed to produce a reasoning block before the final boxed answer.

No-thinking prompt format

Select the single best answer. Respond with exactly one capital letter inside `\boxed{}` and nothing else. No explanation, no option text, no extra words. Output only the boxed letter, for example `\boxed{B}`. `/no_think` Question: {question_text}
Choices: {choices}
Expected assistant output:

```
\boxed{B}
```

Thinking prompt format

You are a careful multiple-choice general-knowledge assistant. Always show your reasoning step by step inside the `<think>` block before giving the final answer. Then finish with exactly one capital letter inside `\boxed{}`. Do not answer without explicit reasoning.
Question: {question_text}
Choices: {choices}
Expected assistant output:

```
<think>
Reasoning steps ...
</think>
\boxed{B}
```

B.3 Main SFT and Inference Experiments

Table 6 summarizes the main General Knowledge SFT and inference experiments. The no-thinking base model achieved only ∼8% local accuracy because many outputs were missing or malformed. Enabling thinking on the base model improved local accuracy to 48.4%, showing that Qwen3-1.7B already had useful reasoning ability, but did not fully solve boxed-answer compliance.

Among no-thinking SFT models, Exp. 1 provided the best balance between local accuracy and CI generalization. Exp. 2 slightly improved local accuracy after adding Quartz v1 and ARC-Challenge, but CI dropped sharply from 44% to 34%. We hypothesize that this was caused by distribution shift, since Quartz contains many binary A/B questions and may have shifted the answer distribution away from the hidden benchmark. Exp. 3 further scaled the data to roughly 126k examples and achieved the strongest no-thinking local score, 49.90%, but its CI score remained lower than Exp. 1. This suggests that larger mixtures improved local performance but did not necessarily transfer better to CI. The final model, Exp. 4, enabled thinking-mode inference on top of Exp. 1 and achieved the best overall result.

Exp.	Training / inference setup	Size	Local	CI
0a	Base Qwen3-1.7B with no-thinking prompt	–	7.51%	–
0b	Base Qwen3-1.7B with thinking enabled	–	48.4%	–
1	MMLU-aux 25k + Nemotron-RL Knowledge MCQA 25k + GPQA extended 546 + SuperGPQA 10k, no-thinking format	∼60k	47.62%	44%
2	Exp. 1 + Quartz v1 2.6k + ARC-Challenge 700, no-thinking format	∼64k	48.16%	34%
3	MMLU-aux 50k + Nemotron-RL Knowledge MCQA 50k + GPQA extended 546 + SuperGPQA 26k, no-thinking format	∼126k	49.90%	37%
4	**Thinking-mode inference on top of Exp. 1**	∼60k	**53.23%**	**45%**

Table 6: Main General Knowledge SFT and inference experiments. Exp. 4 is the final submitted General Knowledge model.

B.4 Additional Scaling Ablations

Table 7 reports smaller-scale and intermediate SFT experiments. These runs were used to identify useful mixtures before training the stronger models in Table 6. They show that moving from small curated datasets to broader academic mixtures substantially improved local accuracy, but later experiments also showed that larger mixtures did not always transfer better to CI. This supports the conclusion that data quality and benchmark alignment were as important as raw dataset size.

Exp.	Training mixture	Size	Local eval	CI eval
A0	Base Qwen3-1.7B with no-thinking prompt	–	7.51%	–
A1	MMLU-aux 6k + ARC-Challenge 750 + OpenBookQA 250 + SciQ 1k	∼8k	42.19%	25%
A2	MMLU-aux 4k + SciQ 1k + GPQA extended	∼5k+	40.85%	31%
A3	MMLU-aux 50k + Nemotron-RL Knowledge MCQA 50k + GPQA extended 546	∼100k	44.78%	34%

Table 7: Additional General Knowledge ablations evaluated on the fixed 2,010-example local evaluation set. These experiments were used to study the effect of scaling, dataset diversity, and no-thinking formatting before selecting the final model.

B.5 GRPO Refinement

We tested GRPO on hard GPQA and SuperGPQA examples using the large no-thinking SFT checkpoint from Exp. 3. Since Exp. 3 had the strongest no-thinking local score but weaker CI performance, the goal was to test whether reward optimization could improve answer selection while preserving boxed-answer formatting. GRPO produced only a small local gain and reduced CI performance, suggesting overfitting to the hard-example training distribution. It was therefore not selected for the final model.

Exp.	Setup	Local	CI
G1	GRPO refinement on hard GPQA/SuperGPQA examples, initialized from Exp. 3 in Table 6	51%	36%

Table 8: GRPO refinement for General Knowledge.

B.6 Thinking-Mode and Chain-of-Thought Experiments

Table 9 compares the thinking-mode and CoT variants. The base thinking model was much stronger than the no-thinking base model, confirming that Qwen3-1.7B already had useful reasoning ability. However, it still sometimes failed to end with a valid boxed answer.

For T1, we performed filtered CoT distillation from a Qwen3-8B teacher using MMLU-aux, GPQA, and SuperGPQA examples. We kept only generations where the teacher produced the correct answer and followed the required `\boxed{X}` format. This improved over the no-thinking base and reached 50.15% local accuracy, but CI remained lower than the final model at 39%. For T2, we upscaled this data with approximately 30k Llama-Nemotron science MCQ examples containing DeepSeek-R1 rationales. This reduced local accuracy to 49.10%, likely because the longer external traces made outputs noisier, more repetitive, and less reliable at ending with the final boxed answer.

The best result came from T3, where we enabled thinking at inference time on top of the selected format-aligned SFT checkpoint from Exp. 1. This reached 53.23% locally and 45% on CI, outperforming both teacher-distilled CoT training and the base thinking baseline.

Exp.	Setup	Size	Local	CI
T0	Base Qwen3-1.7B with thinking enabled	–	48.4%	–
T1	Filtered synthetic CoT distillation from MMLU-aux, GPQA, and SuperGPQA using a Qwen3-8B teacher	∼40k	50.15%	39%
T2	T1 data upscaled with approximately 30k Llama-Nemotron science MCQ examples with DeepSeek-R1 rationales	∼70k	49.10%	–
T3	**Thinking-mode inference on top of Exp. 1**	∼60k	**53.23%**	**45%**

Table 9: Thinking-mode and chain-of-thought experiments for General Knowledge. T3 corresponds to the final submitted model.

B.7 Source and Category Breakdown

Figures 3 and 4 report the accuracy breakdown of the best General Knowledge model by dataset source and subject category. The model performs best on Nemotron and MMLU examples, while performance is lower on harder sources such as MMLU-Pro and SuperGPQA. At the category level, performance is stronger on medicine, biology, psychology, and economics, but weaker on professional law, mathematics, chemistry, and law. This suggests that the final model is strongest on standard academic and biomedical-style MCQs, while harder expert-level reasoning and law-heavy questions remain challenging.

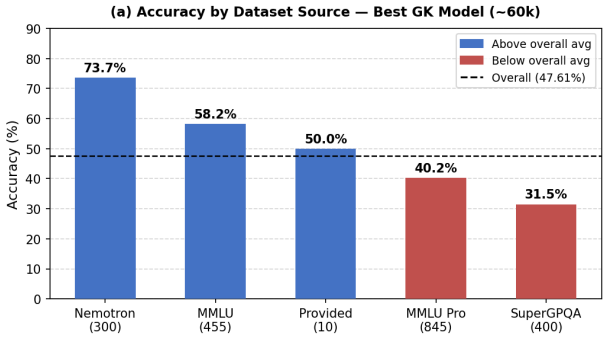


Figure 3: Accuracy breakdown of the best General Knowledge model by dataset source.

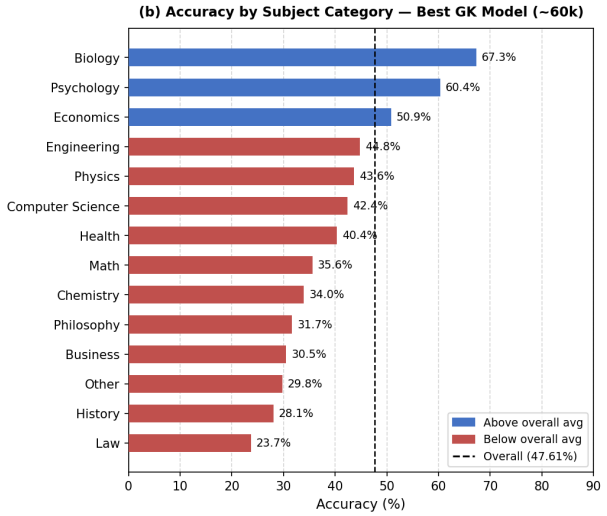


Figure 4: Accuracy breakdown of the best General Knowledge model by subject category.

C Multilinguality Appendix

C.1 data

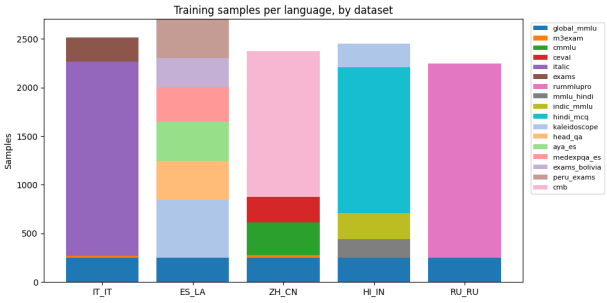


Figure 5: composition of each language from different datasets.

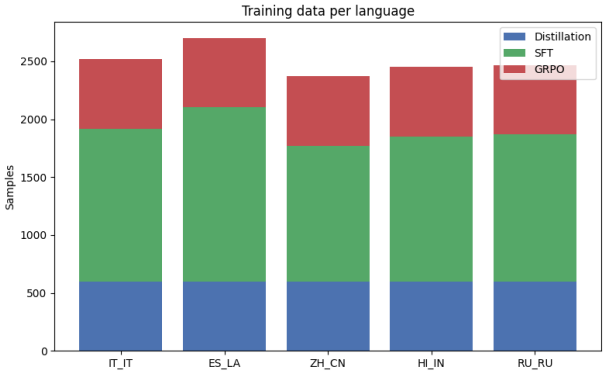


Figure 6: data allocated for each task part of the pipeline (distillation, SFT, GRPO), distillation and GRPO are allocated each 600 samples per language (if present in the pipeline), then SFT takes the rest of the samples.

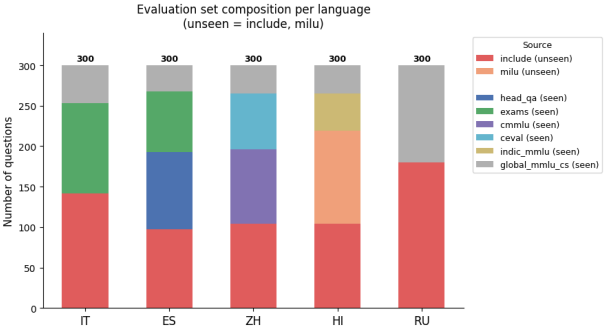


Figure 7: evaluation set composition by dataset and language, datasets contributions are weighted by novelty from the training set (unseen), their subject alignment and their monolinguality.

C.2 Prompts

Multilingual SFT prompt format

```
"sft_system": "Answer multiple-choice
    questions. State the correct option
    and end with \\boxed{LETTER}, where
    LETTER is the correct option (A, B, C,
    etc.)."
```

Multilingual inference prompt format

```
"inference_system": "You are a
    knowledgeable multilingual assistant.
    Read the multiple-choice question
    carefully. Guess an answer using your
    knowledge in English, then select the
    option that best matches your guess
    and is most truthful. Your final
    answer must be a single capital letter
    inside \\boxed{}, for example
    \\boxed{A}."
```

C.3 models and results

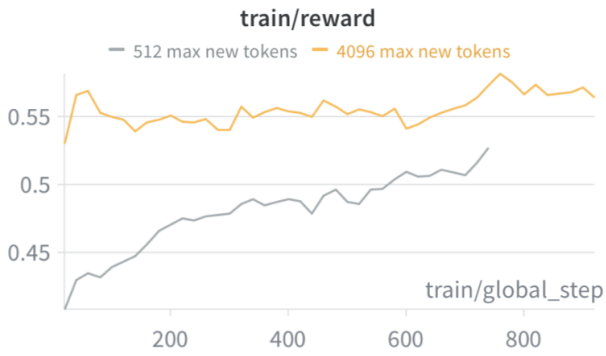


Figure 8: GRPO post SFT trained with 2 different context lenghts, The figure reports that limiting context length shows limited accuracy at first due to format failures, then catches up to a close level to 4k context at the end of the training.

Model	t-off Acc.	f rate	t-on Acc.	f rate
Baseline (Qwen3-1.7B)	49.6%	1.0%	53.9%	2.4%
SFT-mini	44.7%	0.0%	53.9%	2.7%
SFT	50.9%	0.0%	54.9%	1.0%
Distillation-SFT	50.7%	0.1%	54.3%	1.1%
Cold-GRPO	49.7%	0.7%	54.9%	2.1%
SFT-GRPO-mini			55%	0.9%
SFT-GRPO	51.6%	0.0%	54.7%	1.1%
Distil-SFT-GRPO	50.1%	0.0%	56.1%	0.6%
Distil			54.2%	2.4%
Distil-local-lang			52.2%	5.9%

Table 10: Multilinguality experiments evaluated on the 1,500-example local validation set (300 per language). t-off/t-on: thinking disabled/enabled at inference; f rate: boxed-answer format failure rate. GRPO-mini: context length reduced from 4k to 1k in the training

Model	Seen	Unseen	Difference
Baseline (Qwen3-1.7B)	52.4%	56.6%	+4.2%
Cold-GRPO	53.3%	57.5%	+4.2%
Distil-SFT-GRPO	54.2%	57.9%	+2.7%
SFT	53.4%	56.4%	+3.0%

Table 11: The difference in validation between **seen** datasets, which training splits were used for training, and **unseen** which did not have training splits and therefore could have coverage. Unseen accuracy consistently exceeds seen accuracy by 3-4%, suggesting the pipelines generalize beyond training subjects.

Model	ES-LA	HI-IN	IT-IT	RU-RU	ZH-CN
Baseline	53.0%	46.0%	56.7%	53.3%	63.3%
Cold-GRPO	53.3%	45.7%	56.7%	56.0%	62.7%
Distil-SFT-GRPO	55.3%	52.5%	54.0%	53.7%	63.7%
SFT-GRPO	54.7%	46.3%	56.3%	54.7%	63.0%

Table 12: The performance of the top models on different languages. Performance is consistently highest for Chinese (ZH-CN) and lowest for Hindi (HI-IN) across all pipelines.

Exp.	total samples	samples kept	%kept
English distillation	2870	2001	72%
question-language distillation	2870	684	24.6%

Table 13: Difference on performance of the teacher model Qwen3-14B per language, the kept samples are the ones where the teacher model thinks in the right language (english or thespecific language of the question) and succeeds to guess the right answer.

D Safety Appendix

D.1 Prompt Formats

For Safety, we used two target formats depending on whether thinking mode was enabled.

No-thinking safety prompt format

Choose the correct option. Answer only with one boxed letter, like \boxed{A}. /no_think

Question: {question_text}
Choices: {choices}

Expected output: \boxed{B}

Thinking safety prompt format

```
You are a careful reasoning assistant.
Reason through the problem when needed.
Your final answer must appear exactly
once inside \boxed{...}.

Rules: output only the correct capital
letter inside \boxed{...}; do not explain
inside the box.
```

Question: {question_text}
Choices: {choices}

Expected output:
<think> Reasoning steps ...</think>
\boxed{B}

D.2 Evaluation Set Composition

Figure 9 shows the composition of the 935-example held-out evaluation set across eight task types. Safety MCQs dominate (53.5%), providing broad coverage of general harm avoidance, while social norms (16.0%) and toxicity comparison (10.7%) cover complementary aspects of safe behaviour. Minority categories such as bias detection (2.1%), ethics judgement (1.3%), and official course samples (1.1%) ensure coverage of harder edge cases.

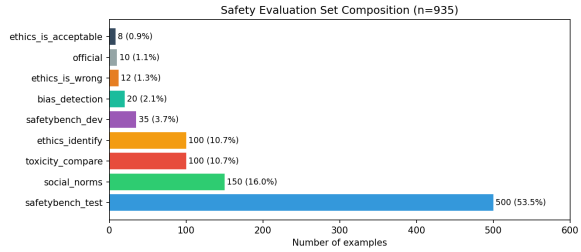


Figure 9: Composition of the 935-example Safety evaluation set across eight task types.

D.3 Accuracy by Task Type

Figure 10 breaks down accuracy by task type for the best Safety SFT model (84% CI, 76.9% overall). Toxicity comparison (92.0%) and bias detection (90.0%) are the strongest categories, likely because they involve clearer surface-level distinctions. Ethics acceptability (37.5%) and ethics identification (63.0%) are the hardest, requiring nuanced moral reasoning that goes beyond surface-level harm classification.

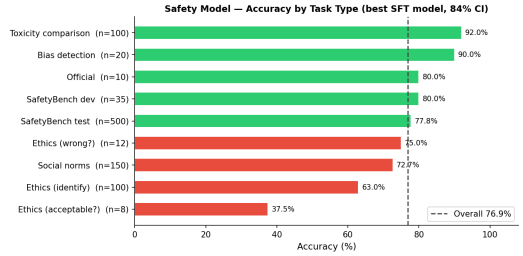


Figure 10: Accuracy by task type for the best Safety SFT model (84% CI, 76.9% overall). Green bars exceed the overall average; red bars fall below.

D.4 SFT Ablations

Table 14 reports all supervised fine-tuning experiments for Safety. The clean 32k mixture (Exp. 3) achieved the best CI score of 84%, outperforming both the WildGuardMix-augmented model (83%) and the larger 58.8k mixture (76%). This suggests the hidden benchmark required broader ethical, bias, and social-reasoning coverage rather than high-quality harm-classification data alone. Scaling further hurt performance as the BBQ-heavy mixture diluted the safety-reasoning signal.

Exp.	Main idea	Size	Clean	CI
1	BeaverTails-only SFT	5.5k	67.70%	–
2	Mixed SFT: BT + SIQA + ETHICS + Civil	10.5k	75.83%	80%
3	Clean mix: BT + HH-RLHF + SIQA + ETHICS + Civil + BBQ + SALAD	32k	77.22%	84%
4	Clean mix + WildGuardMix	48k	78.82%	83%
5	Larger mix with more data and more BBQ	58.8k	74.4%	76%

Table 14: Supervised fine-tuning ablations for Safety

D.5 DPO Experiments

We tested DPO on top of Exp. 3, our strongest clean CI model. Preference pairs were built from hard negatives: examples where the SFT model predicted the wrong option, with the gold boxed answer as chosen and the wrong prediction as rejected. Table 15 reports the results. Although D1 marginally improved clean accuracy, it slightly hurt CI (84% → 83%). Iterative DPO caused severe format collapse: many outputs stopped following the required `\boxed{X}` format. This reflects a fundamental fragility of RL-style refinement on MCQ tasks with near-deterministic single-letter output spaces, where preference updates can over-concentrate the distribution and damage robust format-following.

Exp.	Setup	Clean	CI
D1	One DPO pass on Exp. 3 hard negatives	82.14%	83%
D2	Iterative DPO on D1, 2 epochs	51.98%	–
D3	Softer iterative DPO on D1	53.80%	–

Table 15: DPO experiments after Safety Exp. 3.

D.6 Thinking-mode and CoT Experiments

Table 16 reports thinking-mode and CoT variants for Safety. All experiments used `max_new_tokens = 4096` to avoid truncation of reasoning traces before the final boxed answer. Simply enabling thinking mode (T1) hurt performance when training data contained only boxed answers. Synthetic templated CoT (T2) produced traces that mostly listed options and restated the answer rather than providing genuine reasoning, degrading performance further due to a train/inference thinking-mode mismatch. For real teacher distillation (T3), we compared three teachers: Qwen3-8B (∼2,000 ex/h, lower agreement), Qwen3-14B (∼300 ex/h, better reasoning but too slow), and Claude Haiku 4.5 (∼4,255 ex/h, ∼85% keep rate, best speed–quality trade-off). Despite the higher trace quality, the final 18k CoT model reached only 72.51% on Clean Eval v2, underperforming direct SFT, as the longer reasoning format made optimization harder.

Exp.	Main idea	Size	Clean	CI
T1	Thinking enabled without reasoning traces	32k	75.61%	–
T2	Synthetic templated CoT	38k	70.1%	67%
T3	Real teacher-generated CoT (Claude Haiku 4.5)	18k	72.51%	76%

Table 16: Thinking-mode and CoT experiments for Safety

E Additional Group Model Analysis

E.1 Prompt Templates

For the group baseline, we used a minimal unified prompt to evaluate Qwen3-1.7B without additional task-specific instructions:

Group baseline prompt

```
Answer the question. Always put your final
    answer inside \boxed{}.
For multiple-choice questions, put only the
    option letter inside the box,
for example \boxed{A}.
```

The merged group model used a unified prompt template designed to support all four benchmark types while enforcing a single output format. The template instructed the model to reason inside `<think>...</think>` and to end every response with exactly one final answer inside `\boxed`.

The actual template also included few-shot examples from math, general knowledge, multilinguality, and safety to encourage consistent cross-domain formatting.

Group model prompt

```
You are a careful reasoning assistant
    optimized for four benchmark types:
mathematics, general knowledge,
    multilingual questions, and safety
    questions.

Core behavior:

* Read the full question and all answer
    choices before deciding.
* First identify the task type: math,
    knowledge, multilingual, safety,
  or mixed reasoning.
* For multiple-choice questions, choose the
    single best answer from the
  provided options.
* For math questions, solve step by step,
    verify arithmetic, and give the
  exact final result when possible.
* For general knowledge questions, rely on
    established facts and avoid
  guessing from irrelevant details.
* For multilingual questions, understand
    the question in its original
  language and reason accordingly.
* For safety questions, choose the safest,
    least harmful, most responsible,
  and most policy-compliant option.
* If choices are provided, never invent a
    new answer outside the choices.

Output format:

* Always reason inside <think> ... </think>.
* Always finish with exactly one final
    answer inside one LaTeX \boxed{...}.
* If answer choices are provided, put only
    the option letter inside the box,
  for example \boxed{C}.
* If the task is free-response math, put
    only the exact mathematical answer
  inside the box.
* Do not put explanations, units, or extra
    words inside the box unless
  explicitly required.
* Do not write anything after the boxed
    answer.
```

E.2 Multi-task Training Experiments

Before using adapter merging, we tested whether a single model could be trained directly on data from all four domains. Table 17 reports two multi-task baselines: a direct SFT model and a distillation-SFT-GRPO pipeline trained on the merged datasets from all domains.

Both multi-task approaches underperformed the prompted group baseline and the final linear adapter merge. Direct SFT struggled because mixing Math, General Knowledge, Multilinguality, and Safety examples forced one adapter to learn heterogeneous answer styles, reasoning lengths, and domain-specific behaviours at the same time. Although the distillation-SFT-GRPO pipeline improved over direct SFT, it remained below the linear merge, suggesting that reward refinement did not remove the underlying cross-domain interference. In particular, optimizing a shared model for both open-ended Math answers and single-letter MCQ outputs made formatting and reasoning objectives compete. These results motivated the shift to post-hoc adapter merging, where each domain-specific adapter could first specialize independently before being combined.

Experiment	Local Eval	CI Eval
Multi-task SFT	37.8%	47.5%
Distillation-SFT-GRPO	48.1%	49.2%

Table 17: Group-model multi-task training experiments before adapter merging.

E.3 Merge Strategy Comparison

Before selecting the direct weighted linear merge, we attempted TIES-merging, which resolves sign conflicts and trims redundant parameters across adapters. However, our adapters had significantly different LoRA ranks ($r = 16$ for Safety, $r = 32$ for General Knowledge, and $r = 64$ for Multilinguality and Math) and different training scales, making trimming and sign-election unreliable. Parameters that appeared redundant under TIES were often domain-critical for lower-rank adapters, leading to performance degradation across all domains.

The linear merge avoided this by preserving all adapter updates unconditionally, summing the weighted deltas directly onto the base Qwen3-1.7B weights:

$$\theta_{\text{group}} = \theta_{\text{base}} + \sum_{d \in \mathcal{D}} \lambda_d \cdot \Delta\theta_d \tag{1}$$

where $\mathcal{D}$ = Safety, GK, Multilingual, Math and $\lambda_d = 0.25$ for all domains in the best run. Equal weights were selected after observing that domain-specific tuning of λ_d did not consistently improve CI performance, likely because the hidden benchmark distribution was unknown at merge time. Table 18 compares TIES against the best direct linear merge on the same 2,000-example local evaluation set. TIES reached 54.6% overall, underperforming the linear merge by 3.3 points. The largest drop was in General Knowledge (-8.4 pp), suggesting that heterogeneous adapter ranks and training scales made TIES too aggressive for this setting.

Domain	**TIES Score**	**vs. Linear**
Safety	73.6%	-2.4 pp
General Knowledge	50.8%	-8.4 pp
Multilinguality	47.0%	-2.2 pp
Math	47.2%	-0.6 pp
Overall	**54.6%**	-3.3 pp

Table 18: TIES merge accuracy on the 2,000-example local evaluation set, with the gap relative to the best direct linear merge.

E.4 Group Model Performance

Figure 11 shows the per-domain accuracy of the best group model, obtained with the direct weighted linear merge, on the 2,000-example local evaluation set. Safety is the strongest domain (76.0%), followed by General Knowledge (59.2%), both above the overall average of 57.9%. Math (47.8%) and Multilinguality (49.2%) fall below the average, reflecting cross-domain interference after merging.

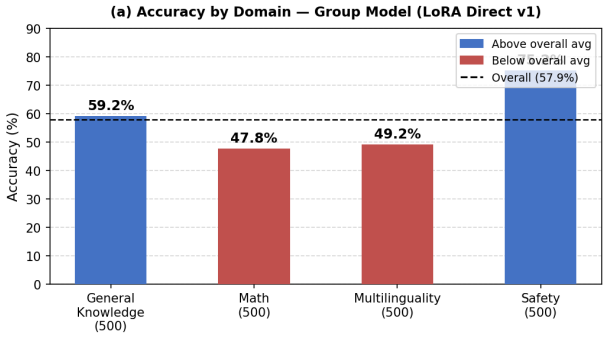


Figure 11: Accuracy by domain for the best group model on the 2,000-example local group evaluation set. The dashed line shows the overall accuracy of 57.9%.

Figure 12 gives a finer sub-source breakdown. The merge preserved performance on standard or well-aligned subsets such as MMLU, MATH-500, SafetyBench, Nemotron, and toxicity comparison, but degraded on harder or more specialized subsets such as AIME, SuperGPQA, EXAMS, and ethics acceptability. AIME requires deep multi-step symbolic reasoning, while EXAMS and SuperGPQA require language- or discipline-specific knowledge that may conflict with broader factual representations learned by the General Knowledge adapter. Ethics acceptability remains the weakest safety sub-domain (35.5%), suggesting that future group models may benefit from domain-adaptive merge weights or post-merge fine-tuning on the weakest sub-domains.

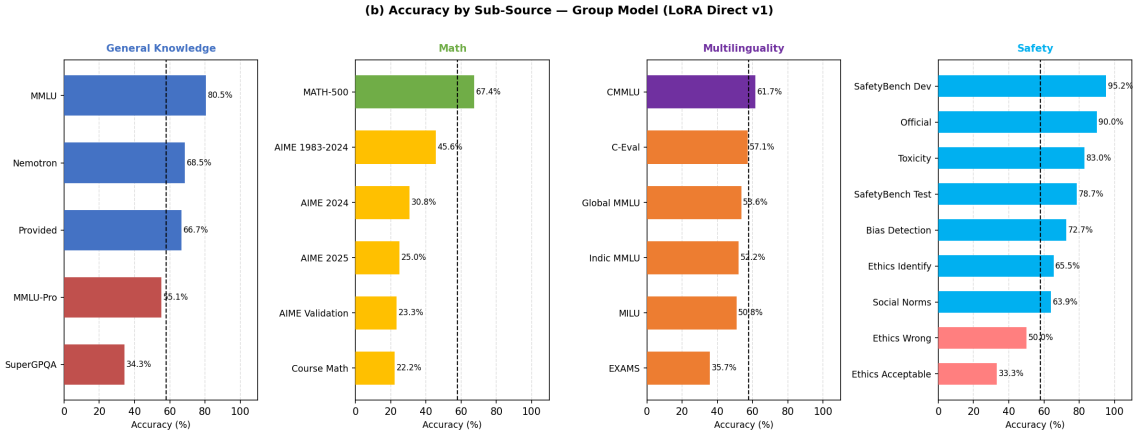


Figure 12: Sub-source accuracy breakdown of the best group model on the 2,000-example local group evaluation set. The dashed line indicates the overall accuracy of 57.9%.